

Hybrid Wafer Map Reconstruction from 9-Site Data

Hybrid Wafer Map Reconstruction from 9-Site E-Test Data

1. Purpose

This document captures and formalizes our discussion on how to reconstruct a **full wafer map** when only **9-site real measurements** are available, by combining:

- **Real 9-site wafer data** (anchor / truth)
- **A representative sample full-wafer distribution** (spatial shape prior)
- **Hard enforcement of wafer median preservation**

The goal is to enable **physically realistic wafer maps** for visualization, correlation, and downstream modeling (e.g., PPV, Vmin, AI/IDEAL flows) while remaining statistically and methodologically defensible.

2. Problem Context

In many Intel/TSMC data flows, E-test or inline parameters are measured at **9 sites per wafer**. This provides:

- Good wafer-to-wafer magnitude information
- Very limited within-wafer spatial resolution

At the same time, other datasets (golden wafers, dense E-test runs, historical lots) may contain **nearly full wafer coverage**, capturing realistic spatial behavior such as:

- Center-edge signatures
- Radial gradients
- Scanner or dosing tilt

Using either source alone is insufficient:

- **Pure 9-site interpolation** → noisy, unrealistically smooth, sensitive to site placement
 - **Pure sample wafer reuse** → ignores real wafer magnitude and biases median
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3. Design Principles

The adopted solution is guided by four principles:

1. **Separate spatial shape from magnitude**
2. **Anchor magnitude using real measurements only**
3. **Borrow spatial structure from trusted reference wafers**
4. **Enforce wafer-level statistics explicitly (median)**

This keeps the method explainable and review-safe.

4. Mathematical Model (Conceptual)

The final reconstructed wafer map is defined as:

$$\text{Final Map} = \alpha \cdot (\text{IDW from 9 sites}) + (1-\alpha) \cdot (\text{Scaled Sample Shape})$$

with a final adjustment to enforce:

$$\text{Median}(\text{Final Map}) = \text{Median}(\text{real 9-site wafer data})$$

Where:

- α is a tunable blending factor (0–1)
 - All spatial shape information is zero-median prior to scaling
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5. Step-by-Step Methodology

Step 1 – Sample Wafer Shape Normalization

From a dense sample wafer map:

```
1 SampleShape(x,y) = SampleValue(x,y) - Median(SampleValue)
2
```

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This removes magnitude and retains only spatial pattern.

Step 2 – 9-Site Interpolation (IDW)

For each location (x,y), compute inverse-distance weighted interpolation using the 9 measured sites:

```
1 V_IDW(x,y) = Σ [ Vi / d_i^p ] / Σ [ 1 / d_i^p ]
2
```

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typically with $p = 2$.

Step 3 – Sample Shape Scaling Using Real Sites

At the 9 site locations, compare the real values to the sample shape:

```
1 ScaleFactor_i = (V_site_i - RealMedian) / (SampleShape_i + ε)
2
```

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Use the **median scale factor** across the 9 sites for robustness:

```
1 WaferScale = Median(ScaleFactor_i)
2
```

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Step 4 – Hybrid Map Construction

```
1 SampleScaled(x,y) = RealMedian + WaferScale * SampleShape(x,y)
2
3 Hybrid(x,y) =  $\alpha$  * V_IDW(x,y) + (1 -  $\alpha$ ) * SampleScaled(x,y)
4
```

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This blends real-data interpolation with physically realistic shape.

Step 5 – Median Enforcement (Hard Constraint)

```
1 FinalMap = Hybrid + (RealMedian - Median(Hybrid))
```

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This guarantees wafer-level fidelity regardless of blending weight.

6. Implementation

A complete **single-file JSL implementation** was developed that:

- Supports multiple wafers
- Uses nearest-neighbor lookup for sample shape anchoring

- Builds IDW, sample-scaled, and final maps explicitly
- Enforces the wafer median exactly

The script outputs a combined table with:

- IDW_Map
- SampleScaled_Map
- Final_Map

This allows easy diagnostics and validation.

7. Validation & Sanity Checks

Recommended checks after reconstruction:

1. **Median preservation**
 - `Median(Final_Map) == Median(9-site data)`
 2. **Site consistency**
 - Final map values near site locations agree with real measurements
 3. **Distribution realism**
 - Histogram shape resembles sample wafer
 - Wafer-to-wafer shifts follow real site data
 4. **Correlation stability**
 - Improved correlation vs wafer-average only methods
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8. Appropriate Use Cases

✓ Suitable for:

- WAT / BEOL resistance
- Ring / Vmin correlation studies
- PPV screening
- Visualization and ML feature generation

⚠ Not suitable for:

- Localized defect mapping
 - Particle scratches
 - One-off excursion analysis
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9. Key Takeaway

This hybrid approach provides a **controlled, explainable, and scalable** way to extend 9-site wafer data into full wafer maps without overstating measurement resolution. It aligns with existing E-test practices and supports advanced modeling workflows while maintaining statistical discipline.

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